

Four-Franka Multi-Planner Integration

Prioritized Planning · centralized CRRT/CRRT-EB · KCBS · current VanillaRRT and FlowEBRRT

Outcome: all six planner/base combinations now share the existing four-arm table world, the 295-sphere MorphIt/cuRobo robot model, world-from-base transforms, and one exact collision/audit definition. PyBullet was not used for planning or evaluation.

What was changed

Area	Implementation
Shared geometry	Reused the four cuRobo checkers, table/pedestal/mount cuboids, fixed world-from-base transforms, and 295 world-frame spheres per arm. Added flattened joint-path collision and synchronized first-conflict callbacks.
Prioritized	Reused the existing dynamic-obstacle adapter. Each lower-priority arm checks its cuRobo edge against cached sphere trajectories of higher-priority arms.
CRRT	Added optional articulated metric and exact joint-path callbacks to the legacy centralized planner. Goal-biased parent/extension selection uses each Franka agent's 7D position policy; ordinary parent selection remains 14D.
CRRT-EB / Flow	Added a dynamic centralized Flow variant using the current best_inference.pt model. It generates 32 sequence edges per active arm, ranks by normalized 7D position, physically propagates each sequence, repairs colliding joint candidates, then falls back to random controls.
KCBS	Added an exact conflict-detector callback and exact time-indexed constraint builder. Low-level Vanilla/Flow replanning checks 295-sphere overlaps against the constrained arm state at the matching timestep; legacy point-radius geometry is bypassed.
Budgeting	Enforced one 30 s total budget per four-arm trial. Added pre-extension deadline checks so a zero-remaining-budget RRT/CRRT call cannot execute one expensive rollout. KCBS passes only the remaining budget to every low-level call.
Reproducibility	Created a resumable 360-trial runner. Every trial writes one JSON record; audited successes also store four trajectories. Aggregates are regenerated from raw records.

Scenario calibration and protocol

Tier	Direct invalid	Arm pairs	Mean normalized q motion	Purpose
Easy	0/39	0	0.475	Short, nontrivial; each arm starts outside radius 0.40
Medium	1/39	1	1.806	Longer motion with one exact inter-arm bottleneck
Hard	18/39	2	2.037	Large motion with two exact inter-arm bottlenecks

Protocol: 20 evaluation seeds per tier and planner/base combination; 3 tiers × 6 combinations = 360 trials. Success requires planner completion and a post-run audit of joint/velocity limits, cuRobo self/static collision, and synchronized world-frame inter-arm sphere overlap. Timing statistics below include successful trials only.

Files and reproducibility

Adapters: scripts/FrankaObstacle/four_franka_multi_planner.py and franka_prioritized_adapter.py. Runner: benchmark_four_franka_multi_planner.py. Scenario generator: create_four_franka_benchmark_problems.py. Legacy integration points: mrmp_with_kite_extend/src/cRRT.py, kcbs.py, and rrt.py. Raw JSONs, trajectories, CSVs, summary, and hashes are in results/franka_obstacle/four_franka_multi_planner_20seeds_final_v3_20260812/.

Overall results (60 trials per row)

Coordination	Base	Success	Plan mean / median (s)	Makespan mean / median (s)	Nodes total	Waypoints total
Prioritized	Vanilla	20/60 (33.3%)	0.36 / 0.33	0.31 / 0.29	14,377	2,777,752
Prioritized	Flow	19/60 (31.7%)	13.49 / 12.32	3.90 / 1.58	97,161	3,225,633
Crrt	Vanilla	2/60 (3.3%)	15.04 / 15.04	0.69 / 0.69	14,078	4,574,382
Crrt	Flow	3/60 (5.0%)	22.85 / 26.00	9.39 / 10.22	31,297	3,393,536
Kcbs	Vanilla	20/60 (33.3%)	0.34 / 0.18	0.30 / 0.28	18,819	3,604,154
Kcbs	Flow	13/60 (21.7%)	5.42 / 5.99	1.92 / 1.34	125,426	3,955,625

Per-tier success and successful-trial timing

Coordination + base	Tier	Success	Plan mean / med (s)	Makespan mean / med (s)
Prioritized + Vanilla	Easy	20/20 (100%)	0.36 / 0.33	0.31 / 0.29
Prioritized + Vanilla	Medium	0/20 (0%)	— / —	— / —
Prioritized + Vanilla	Hard	0/20 (0%)	— / —	— / —
Prioritized + Flow	Easy	13/20 (65%)	12.30 / 10.61	1.34 / 1.26
Prioritized + Flow	Medium	3/20 (15%)	14.29 / 12.32	11.13 / 10.02
Prioritized + Flow	Hard	3/20 (15%)	17.87 / 15.98	7.75 / 7.50
CRRT + Vanilla	Easy	2/20 (10%)	15.04 / 15.04	0.69 / 0.69
CRRT + Vanilla	Medium	0/20 (0%)	— / —	— / —
CRRT + Vanilla	Hard	0/20 (0%)	— / —	— / —
CRRT + Flow	Easy	1/20 (5%)	26.00 / 26.00	1.84 / 1.84
CRRT + Flow	Medium	1/20 (5%)	12.90 / 12.90	10.22 / 10.22
CRRT + Flow	Hard	1/20 (5%)	29.66 / 29.66	16.12 / 16.12
KCBS + Vanilla	Easy	20/20 (100%)	0.34 / 0.18	0.30 / 0.28
KCBS + Vanilla	Medium	0/20 (0%)	— / —	— / —
KCBS + Vanilla	Hard	0/20 (0%)	— / —	— / —
KCBS + Flow	Easy	12/20 (60%)	5.14 / 5.84	1.50 / 1.34
KCBS + Flow	Medium	1/20 (5%)	8.72 / 8.72	6.86 / 6.86
KCBS + Flow	Hard	0/20 (0%)	— / —	— / —

Interpretation: excluded pilots confirmed easy. Vanilla prioritized/KCBS were fastest and 100% reliable there. Flow produced the only medium/hard successes for Prioritized and CRRT, at much larger node/waypoint cost. Centralized CRRT was least reliable within 30 s. All 84 successes passed exact post-run audit.